CHAPTER 4

Describing Our App Declaratively with Docker Compose

We've seen how to start containers using the `docker run` command. However, this is limited to starting a single container at a time. While that's great for one-off tasks, as developers, our applications are typically made up of multiple components, or in Docker terminology, *services*. For example, in addition to our Rails server, we typically need at least a database.

Once you're thinking about an application as a whole, running containers with `docker run` becomes too cumbersome. We need a different, higher-level tool that lets us coordinate and manage containers for the different services that make up our application.

Enter Docker Compose.

Getting Started with Compose

Docker Compose—or just Compose for short—is a tool for managing an application that needs several different containers to work together. Compose is declarative: you describe each part of your application—known as a *service*—and Compose handles the grunt work of ensuring the right containers are run when and how you need. It also manages creating and destroying the resources needed for the app. For example, it creates a separate, private network for your application, giving you a predictable, isolated environment. As we'll see in Part II, Toward Production, on page 127, it also plays a key role in how we deploy and scale applications with Docker.

Compose is designed with developers in mind. It lets us interact with our application in terms of concepts we're familiar with—for example, "run our